

ECEN 607: Advanced Analog Circuits Tech

Lab 5: Two-stage folded-cascode Op Amp with Class AB Output stage

Name : FirstName LastName
UIN : ***Last Three Digits**

Objectives

- To study and design a two-stage op amp with Class AB output stage.
- To study about the linearity in circuits and how to measure them.

This lab is based on the two-stage folded-cascode-class AB amplifier shown below. Transistors M1-M10 implement the amplifier's first stage while transistors M11a-b implement the Monticelli's biasing circuit. For the second stage. Due to the cascode transistors associated with the first stage, it is expected to achieve large gain in the first stage; e.g. over 30dB. The last stage (M16-M17) may achieve moderate gain; e.g. 20dB, however, this stage provides large signal swing and very good current capabilities due to its class AB nature.

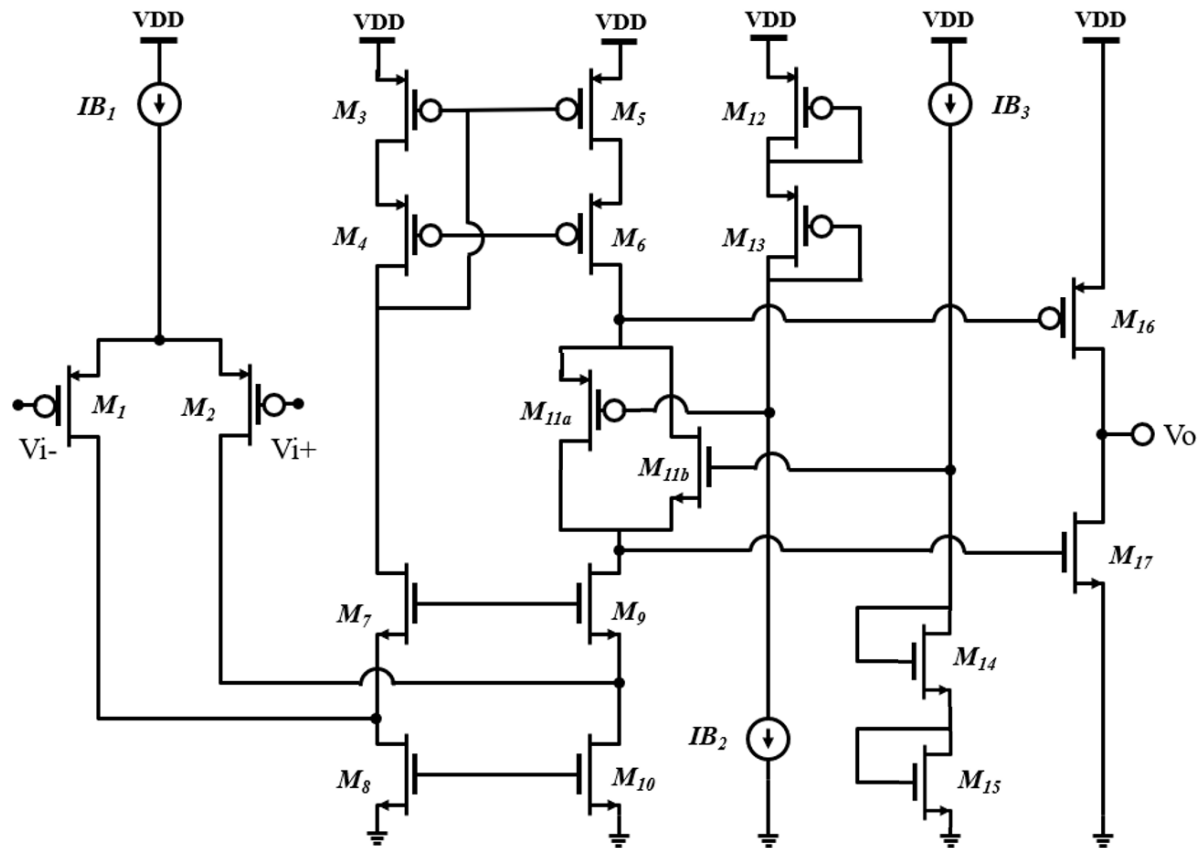


Figure 6-1: Two-stage folded-cascode amplifier with class AB output stage

Prelab

- In this prelab, use 5.5V NFETs and PFETs.
 - Show a detailed design procedure for each circuit.
 - Make a table reporting the bias current and dimensions for each transistor
1. Design the circuit to meet the specs given table 6-1. Add a suitable compensation scheme for the circuit. Explain your reasoning.
 - a. Employ ideal current sources for the ones marked in the schematic (IB1, IB2 and IB3), but bias properly transistors M8 and M10.

- b. The circuit requires phase compensation; Miller compensation could be an option, but also feedforward could be used. Justify the phase compensation methodology you will use.
 - c. Explain the slew-rate mechanism (both positive and negative) with equations. Large input voltage variations are necessary in this case.
2. Identify the dominant noise sources (you don't have to fully derive the transfer functions) and explain your reasoning.

Table 6-1: Design specifications

Supply Voltage range (USB spec)	5V
A_{vo}	> 60 dB
Slew Rate	4 V/us
GBW	> 10 MHz
PM	> 45 deg
Output Swing (differential pk-pk)	4.5 V
C_L	20pF
Quiescent Power (if possible)	< 1mW

Lab Report

1. Simulate the circuit from the prelab and adjust the transistor sizes accordingly until you meet all specifications given in Table 6-1.
 - a. You should also make sure that the THD (3rd harmonic distortion) < 1 % for a single tone of 2V_{pk-pk} input at 50 kHz when the op-amp operates in unity buffer configuration.
 - b. Provide the necessary plots (AC and transient) such that you show that you meet all the specs; e.g. magnitude and phase (AC) simulations,
 - c. Proper transient simulations for the sinusoidal input defined in a) (to measure HD3),
 - d. Pulse response to measure amplifier's slew-rate, and step transient; employ a pulsed signal of 200mV with a frequency of 1KHz. Measure the 1% settling time.
2. Build a unity gain capacitive inverting amplifier with two 1pF capacitors. You may want to add a 10Mohm resistor in feedback to stabilize the operating point.
 - a. Apply a single tone of 2V_{pk-pk} input at 50 kHz. For THD; run transient simulations and show the PSD of the output and mark the harmonic powers (up to 7th one).
 - b. Sweep the amplitude of the input signal in steps of 200mV_{pk-pk} until 2V_{pk-pk} with steps of 200mV_{pk-pk}. and make a plot showing HD2 and HD3 versus input signal amplitude. Cadence Spectrum tool calculates the THD when number of harmonics to be considered is specified. Make sure that i) the strobe period for the transient run is given properly and ii) you pick the points that are present in the transient run instead of the linearly approximated value (else THD numbers could be wrong). Use rectangular window only. Mention your choice of relevant parameters in the report.
 - c. Plot the HD3 when sweeping the frequency from 10kHz till 10MHz; use frequency decades (1k, 10k, 100k, etc.). Fix the amplitude of the input signal to 3V peak.
 - d. Include screenshots of FFT settings that you used, THD results and comment on the results.
 - a. What are the major sources of non-linearities in your circuit and suggest ways to improve linearity. Mention the trade-offs if any.
3. Provide comparison tables of hand-calculated vs. final transistor sizes, and required specs vs. simulated specs.

4. Measure the -1dB voltage gain compression point. Use PSS approach or transient V_{out_peak}/V_{in_peak} . (Show relevant plots and report the numbers.)
5. Comment on your results and provide conclusions.

Reference

- [1] R. Hogervorst, J. P. Tero, R. G. H. Eschauzier and J. H. Huijsing, "A compact power-efficient 3 V CMOS rail-to-rail input/output operational amplifier for VLSI cell libraries," in *IEEE Journal of Solid-State Circuits*, vol. 29, no. 12, pp. 1505-1513, Dec. 1994.
- [2] D. M. Monticelli, "A quad CMOS single-supply op amp with rail-to-rail output swing," in *IEEE Journal of Solid-State Circuits*, vol. 21, no. 6, pp. 1026-1034, Dec. 1986, doi: 10.1109/JSSC.1986.1052645

Rev. History

2025.03.04	Figure update: MOSFET symbols
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